

HDOC 12

Activity Description (Batches 15 & 16):

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum and product of the first and last digits. E.g. Number: 12345, Sum = 6, Product = 5
- Count how many digits are lesser than, equal, and greater than a given digit k. E.g. Number: 12345, k=3, Count_lesser=2, Count_equal=1, Count_greater=2

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1. Algorithm

1. Start

2. Declare variables for choice, num, k, temp, rem, first_digit, last_digit, sum, prod, count_less, count_equal, count_greater.
3. Display the menu options to the user.
4. Read the user's choice.
5. If choice is 1 or 2, prompt the user to input a positive integer num. Validate that $\text{num} > 0$.
6. Use a switch statement based on the value of choice.

7. Case 1 (Sum and Product of first and last digits):

- Calculate $\text{last_digit} = \text{num} \% 10$.
- Set $\text{temp} = \text{num}$.
- Loop while ($\text{temp} \geq 10$): set $\text{temp} = \text{temp} / 10$.
- After the loop, set $\text{first_digit} = \text{temp}$.
- Calculate $\text{sum} = \text{first_digit} + \text{last_digit}$ and $\text{prod} = \text{first_digit} * \text{last_digit}$.
- Display sum and prod, then break.

8. Case 2 (Count digits compared to k):

- Prompt the user to input a digit k.
- Initialize $\text{count_less} = 0$, $\text{count_equal} = 0$, $\text{count_greater} = 0$.

- Set temp = num.
- Loop while (temp > 0):
 - Calculate rem = temp % 10.
 - If rem < k, increment count_less.
 - Else if rem == k, increment count_equal.
 - Else, increment count_greater.
 - Update temp = temp / 10.
- Display count_less, count_equal, and count_greater, then break.

9. **Default Case:** Display an "Invalid choice" error message.

10. **Stop**

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num, k)	Expected Output
TC-01	Valid inputs for Menu Option 1	Choice: 1 Num: 12345	Sum = 6 Product = 5
TC-02	Valid inputs for Menu Option 2	Choice: 2 Num: 12345 k: 3	Count_lesser: 2 Count_equal: 1 Count_greater: 2
TC-03	Single digit number (Option 1)	Choice: 1 Num: 7	Sum = 14 Product = 49
TC-04	All digits same as k (Option 2)	Choice: 2 Num: 5555 k: 5	Count_lesser: 0 Count_equal: 4 Count_greater: 0
TC-05	Invalid negative number input	Choice: 1 Num: -500	Error: Please enter a strictly positive integer.
TC-06	Invalid menu choice	Choice: 4	Invalid choice. Please choose 1 or 2.

3. C Program Source Code

```
#include <stdio.h>

int main() {
    int choice;
    long long num, temp;
    int k, rem, first_digit, last_digit;
    int sum = 0, prod = 1;
    int count_less = 0, count_equal = 0, count_greater = 0;

    // Display Menu
    printf("--- HDOC 12 Operations Menu ---\n");
    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to, and greater than k\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    // Validate and Take Input
    if (choice == 1 || choice == 2) {
        printf("Enter a positive integer: ");
        scanf("%lld", &num);

        if (num <= 0) {
            printf("Error: Please enter a strictly positive integer.\n");
            return 1; // Exit program due to invalid input
        }
    }

    // Switch Case Logic
    switch (choice) {
        case 1:
            last_digit = num % 10;
            temp = num;

            // Loop to find the first digit
            while (temp >= 10) {
                temp /= 10;
            }
            first_digit = temp;

            sum = first_digit + last_digit;
            prod = first_digit * last_digit;

            printf("Number: %lld\n", num);
            printf("Sum of first and last digit: %d\n", sum);
```

```
        printf("Product of first and last digit: %d\n", prod);
        break;

    case 2:
        printf("Enter the digit k (0-9): ");
        scanf("%d", &k);

        temp = num;
        while (temp > 0) {
            rem = temp % 10;
            if (rem < k) {
                count_less++;
            } else if (rem == k) {
                count_equal++;
            } else {
                count_greater++;
            }
            temp /= 10;
        }

        printf("Number: %lld, k: %d\n", num, k);
        printf("Count_lesser: %d\n", count_less);
        printf("Count_equal: %d\n", count_equal);
        printf("Count_greater: %d\n", count_greater);
        break;

    default:
        printf("Invalid choice. Please choose 1 or 2.\n");
        break;
}

return 0;
}
```